

AI & Machine Learning Internship — Task 2

Feature Engineering, Model Optimization & Performance Comparison

Intern:	Khushi Singh
Organization:	Maincrafts Technology
Dataset:	California Housing Dataset (scikit-learn)
Tools:	Python, pandas, NumPy, scikit-learn, Matplotlib, Google Colab

1. Objective

The goal of this task was to move beyond a basic Machine Learning model (Task 1) and follow a professional workflow: prepare data correctly using feature scaling, train multiple regression algorithms, evaluate them using standard metrics, and select the best-performing model with data-driven justification.

2. Methodology

- **Data Loading:** Loaded the California Housing dataset (20,640 records, 8 features) using scikit-learn's `fetch_california_housing()`.
- **Feature/Target Split:** Separated the 8 input features (X) from the target variable, Median House Value (y).
- **Feature Scaling:** Applied `StandardScaler` to normalize all features to a common scale, preventing large-range features (e.g. Population) from dominating smaller-range ones (e.g. AveRooms).
- **Train-Test Split:** Split the data 80/20 (`random_state=42`) to evaluate models on unseen data.
- **Model Training:** Trained three regression models — Linear Regression (baseline), Ridge Regression (`alpha=1.0`, for regularization), and Decision Tree Regressor (`max_depth=5`, to capture non-linear patterns).
- **Evaluation:** Compared all models using RMSE (Root Mean Squared Error) and R^2 Score.
- **Visualization:** Plotted Actual vs Predicted house prices for the best-performing model to visually validate performance.

3. Results

The table below summarizes the performance of all three models on the test set:

Model	RMSE ↓	R^2 Score ↑
Linear Regression	0.7456	0.5758
Ridge Regression	0.7456	0.5758
Decision Tree Regressor	0.7242	0.5997

Best Performing Model: Decision Tree Regressor — selected based on the lowest RMSE (0.7242) and the highest R^2 Score (0.5997) among the three models tested.

4. Key Insights

- Feature scaling was essential: without it, features like Population (range: 3–35,682) would have dominated smaller-scale features like AveRooms (range: 1–10), destabilizing model training.
- Ridge Regression performed almost identically to plain Linear Regression (RMSE 0.7456 for both), indicating that overfitting was not a significant issue for this baseline feature set — regularization had limited

additional benefit.

- The Decision Tree Regressor outperformed both linear models because it could capture non-linear relationships in the data (e.g. complex interactions between location and income) that linear models cannot represent.
- Limiting tree depth (`max_depth=5`) was necessary to prevent the Decision Tree from overfitting to the training data while still allowing it to model non-linear patterns.

5. Conclusion

This task demonstrated a complete, industry-aligned Machine Learning workflow — from data preprocessing and feature scaling to training multiple algorithms and making a data-driven model selection. The Decision Tree Regressor was chosen as the final model because it best captured the non-linear structure of the California Housing dataset, achieving a lower RMSE and a higher R^2 Score than both linear alternatives. This exercise reinforced the importance of comparing multiple approaches rather than relying on a single model, which mirrors how Machine Learning engineers optimize models in real-world projects.